TIC 80518171

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period       7.956676 d
  Epoch (t0)   1576.6838 BTJD
  Duration     0.1700 d
  Depth        16.702 %  (167,016 ppm)
  SNR          78.8

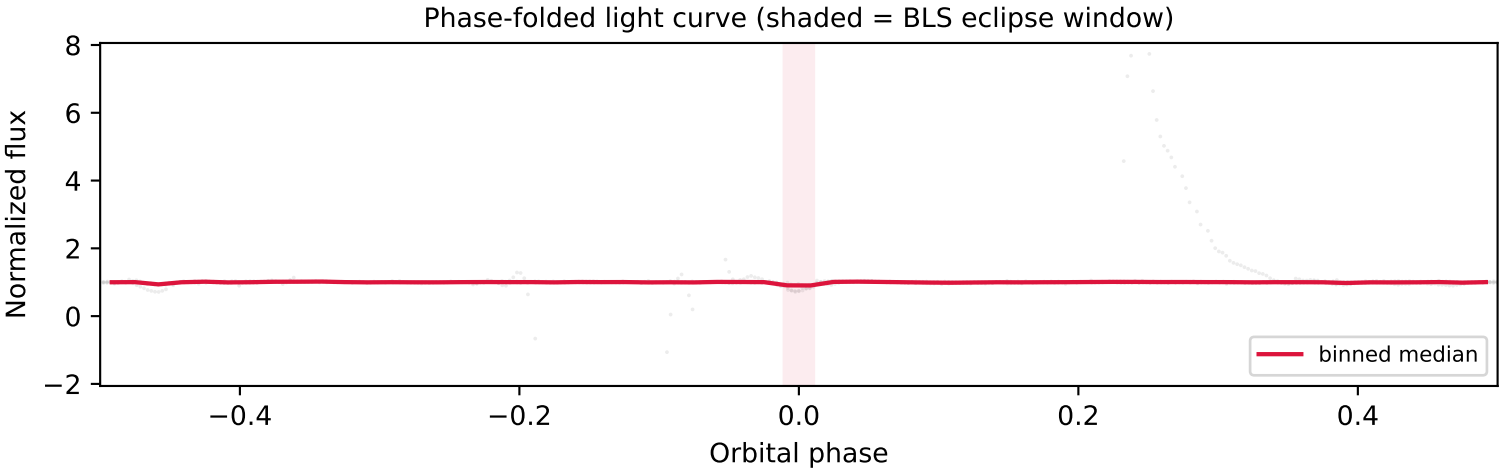
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     3  (gate >= 3)
  Robust depth ratio    1.04  (gate >= 0.50)
  Robust depth sigma    38.1  (gate >= 7.0)
  Odd/even z-score      2.00  (consistent)
  Secondary eclipse     detected, phase -0.46, 5.5 sigma
```

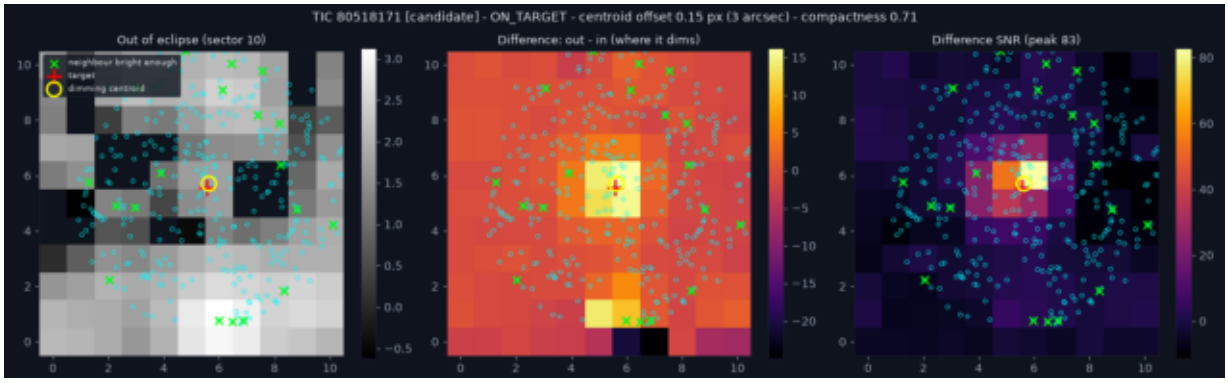
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 10    light-curve dip in this sector: 40.9 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   82.88   (gate >= 5.0)      Compactness   0.71   (gate >= 0.50)
  Dimming centroid offset     0.15 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 80518199
    1.77 px away, dTmag +1.49 - farther from the dimming than the target
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No caveats - clean detection on every check.